

Solihin

Research Assistant, Research Center for Quantum Physics, BRIN
Master's Student, Department of Physics, Universitas Indonesia
Phone: +62 895-1709-3979
Email: solihin@ui.ac.id or solihin1717.wonomulyo@gmail.com
Personal Website: solihin.web.id



Personal Information

Place, Date of Birth: Jogja Lama, Indonesia — January 10, 2003
Address: Bogor Regency, West Java, Indonesia
Languages: English (Advanced), Indonesian (Native), Javanese (Native)

Education

Master of Physics, Universitas Indonesia, Depok, Indonesia (August 2024 – Present). Thesis project: Quantum description of magnon-polariton in electromagnetic cavities.

Bachelor of Science in Physics, Institut Teknologi Sepuluh Nopember, Surabaya, Indonesia (2020 – 2024). Final project: Synthesis and characterization of B/N doped carbon dots.

Work & Internship Experience

Internship Student, Quantum Material Properties Group, Research Center for Materials Nanoarchitectonics, National Research for Materials Science (NIMS), Tsukuba, Japan (August 2026 – Present). Investigate the nematic susceptibility behavior at the coexistence of superconducting and nematic phases.

Physics and Mathematics Teacher, Brain Academy Ruangguru, South Tangerang, Indonesia (January 2024 – Present). Conduct intensive physics and mathematics teaching sessions and academic mentoring for high school students.

Research Assistant, Research Center for Quantum Physics, National Research and Innovation Agency (BRIN), South Tangerang, Indonesia (May 2023 – Present). Investigate the coupling of magnon-photons in electromagnetic cavities and analyze the electronic, topological, and magnonic properties of magnetic materials.

Internship Student, Department of Condensed Matter Physics, Hokkaido University, Japan (November 2025 – December 2025). Studied the anomaly of longitudinal spin susceptibility at superconducting instabilities within magnetic phases and investigated nematic susceptibility in the interplay between nematic and superconducting phases.

Teaching Assistant, Institut Teknologi Sepuluh Nopember, Surabaya, Indonesia (2022 – 2023). Assisted first-year undergraduate students in Fundamental Physics courses, providing guidance in problem-solving and conceptual understanding.

Internship Student, Research Center for Quantum Physics, BRIN, South Tangerang, Indonesia (December 2022 – January 2023). Investigated light propagation in undoped graphene sandwiched between multilayer dielectric structures.

Internship Student, Research Center for Advanced Materials, BRIN, South Tangerang, Indonesia (December 2022 – January 2023). Investigated optical properties of carbon dots synthesized via the microwave and laser ablation method.

Research Interests

Condensed Matter Theory, Superconductivity, Spin Dynamics, Quantum Mechanics.

Publications

3. **S. Solihin**, A. R. T. Nugraha, M. A. Majidi. (2026). Spin Response Properties in Electronically Robust Ferromagnetic Strained CrSiSe₃ Monolayer under External Electric Fields. [*arXiv preprint* arXiv:2605.20021](#).
2. **S. Solihin**, A. R. T. Nugraha, M. A. Majidi. (2026). Geometry-controlled magnon-polariton excitations in a bilayer planar cavity. [*arXiv preprint* arXiv:2604.11690](#).
1. F. Kengo, **S. Solihin**, M. S. Ukhtary, J. Suwardy, M. A. Majidi, A. R. T. Nugraha. (2024). Tunable optical absorption in undoped graphene sandwiched between multilayer dielectric stacks with mirror symmetry. [*Physica Scripta*, 99\(3\), 035513](#).

Technical Skills

Operating Systems: Windows (advanced), Linux (advanced).

Programming: Python (advanced), MATLAB (intermediate), Fortran (basic).

Computational Tools: Quantum ESPRESSO (advanced), Wannier90 (advanced), TB2J (advanced), VESTA (intermediate).

Document Preparation: L^AT_EX(advanced)